

Article

Symmetry-Aware Stock Prediction Based on Optimized Multi-Module Collaborative Features with LSTM-CBAM-Time2Vec-KAN

Huiyong Wu and Xiufeng Hong *

School of Science, Shenyang University of Chemical Technology, Shenyang 110142, China;
wuhuiyong@syuct.edu.cn

* Correspondence: 2024207@stu.syuct.edu.cn

Abstract

This study proposes a hybrid deep learning model named LSTM-CBAM-Time2Vec-KAN based on symmetry awareness and optimized multi-module collaborative features, aiming to improve the accuracy and stability of stock price prediction. To address common shortcomings in traditional forecasting models such as insufficient feature extraction, difficulties in parameter optimization, and inadequate utilization of temporal characteristics, the research innovatively exploits the symmetry inherent in financial time series, particularly their temporal periodicity and cross-dimensional feature consistency, to construct an intelligent prediction framework that integrates multiple modules. First, wavelet transform is applied to perform multi-scale decomposition and signal reconstruction on the raw stock price sequence, effectively extracting high signal-to-noise ratio features. Second, the Northern Goshawk Optimization (NGO) algorithm is employed to jointly optimize key hyperparameters of the model, including the LSTM hidden layer dimension and CBAM compression ratio, thereby resolving the challenge of parameter coupling across modules. Third, the CBAM attention mechanism enhances the importance of temporal features extracted by LSTM through a dual mechanism of channel and spatial attention, enabling the model to focus on critical price movement points. Meanwhile, Time2Vec encoding transforms temporal information into embedding representations with periodic properties, effectively capturing cyclical patterns at daily, weekly, and monthly trading intervals. Finally, the Kolmogorov–Arnold network (KAN) fuses multimodal features and produces precise predictive outputs. Experimental results show that the proposed model significantly outperforms all baseline models in four evaluation metrics, namely mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2), which verifies its superior prediction accuracy and robustness. Furthermore, analyses of stock price forecasting under different time spans and simulated trading performance under various trading strategies further demonstrate that this study provides a feasible and effective technical solution for financial time-series forecasting, with important theoretical research value and practical application value.



Academic Editor: Jie Yang

Received: 26 May 2026

Revised: 4 July 2026

Accepted: 9 July 2026

Published: 16 July 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

Keywords: stock price prediction; attention mechanism; time-series analysis; Northern Goshawk Optimization; multi-module fusion

1. Introduction

Stock market prediction, as a core research topic in financial engineering, faces three major challenges inherent to financial markets: high noise, non-stationarity, and nonlinear

dynamic characteristics. Stock markets and financial variables are notoriously difficult to predict. Accurate forecasting can help individuals, investors, and institutions make better financial decisions, thereby enhancing profitability and mitigating risk. Due to the complexity and diversity of factors influencing the prediction process, stock market forecasting is widely regarded as a challenging research problem. Traditional time-series prediction methods such as ARIMA and GARCH models, which are based on nonlinear assumptions, struggle to capture the complex nonlinear relationships in modern financial markets. With the rise of high-frequency trading and quantitative investing, there are increasing demands for the real-time performance and accuracy of prediction models. While deep learning technologies have achieved breakthroughs in fields like computer vision, their direct application to financial data still faces significant challenges: market microstructure noise interferes with model training, multi-time-scale features are difficult to capture synchronously, and rapid changes in market conditions lead to insufficient model generalization. These characteristics make stock price prediction a typical complex system modeling problem, urgently requiring new methods and technologies for solutions.

Current research on financial time-series forecasting primarily revolves around three directions: improvements of traditional models, design of hybrid architectures, and innovation in optimization algorithms. Through an in-depth analysis of the existing literature, we observed a strong intrinsic relationship between these research advances and the fundamental characteristics of financial time series. In terms of model architecture, CNN-CBAM-LSTM [1] effectively enhances multi-scale feature extraction by integrating convolutional neural networks with attention mechanisms, making it well-suited for capturing both the short-term fluctuations and long-term trends inherent in financial time series. The multi-resolution property of wavelet analysis [2] efficiently handles periodic patterns in the data, and its hybrid approach with ARIMA-LSTM [3] has demonstrated advantages in simultaneously modeling linear trends and nonlinear dynamics, as validated in stock index futures forecasting. Regarding optimization algorithms, bio-inspired methods such as NGO-LSTM [4] and GA-WOA-LSTM [5] accelerate model convergence to better adapt to the structural changes characteristic of financial data. SAO-LSTM [6] exhibits strong generalization capability, reflecting its effectiveness in capturing cyclical fluctuations. Notably, the sustained competitiveness of traditional approaches like CNN-BiLSTM-AM [7] underscores the critical value of bidirectional information flow and attention mechanisms in modeling the long-memory properties of financial time series. Collectively, these studies indicate that high-performing forecasting models require tailored optimization aligned with the specific statistical properties of financial time series. The correspondence between model architecture and data characteristics provides a crucial direction for future research.

Since the advent of the big data era, deep learning-based forecasting models have been regarded as a primary research direction in stock market prediction, demonstrating unique advantages in handling massive time-series data. Representative architectures include recurrent neural networks (RNNs), convolutional neural networks (CNNs) [8], and long short-term memory networks (LSTMs) [9], along with their variants. Some studies systematically sort out the full application context of deep learning in financial forecasting. By summarizing the modeling paradigms of multi-source heterogeneous data, they point out that multimodal feature fusion is a core approach to improving the prediction robustness of models [10]. Other research has focused on multi-step-ahead prediction, validating the effectiveness of deep learning in long-term investment decision-making for major global indices [11]. Collectively, these works have significantly advanced the application of deep learning in financial time-series analysis and provided valuable insights for building more robust forecasting systems. Nevertheless, existing methods still lack explicit modeling of

interactions among variables, and further exploration is needed regarding the definition of prediction horizons and cross-market generalization capabilities.

Recent research has focused on the application of novel network architectures, with Kolmogorov–Arnold networks (KANs) achieving higher parameter efficiency than traditional multilayer perceptrons (MLPs) through the principle of function composition. Studies have demonstrated that KANs offer significant advantages in time-series forecasting. For instance, the C-KAN model [12] innovatively integrates convolutional layers with KAN, enhancing temporal prediction performance. In the domain of electricity demand forecasting, the integration of KAN with conventional time-series forecasting frameworks [13] has shown superior results. The Attention-KAN model [14] further improves prediction accuracy, and KAN has also achieved a breakthrough performance in hyperspectral image classification [15]. These advances provide new perspectives for financial forecasting; however, core challenges such as noise robustness and multi-scale feature fusion have not yet been systematically addressed.

Beyond methodological improvements, the choice of research focus in forecasting is equally critical. Current stock prediction studies predominantly rely on the independent modeling of single-asset price series. While this approach can capture the local dynamics of a specific asset, it suffers from limitations such as difficulty in extracting multi-temporal-scale features and insufficient adaptability to abrupt shifts in market regimes. In reality, financial time series are driven by multiple interrelated factors and are not mutually independent. For example, empirical evidence shows that external shocks—such as oil price fluctuations—significantly influence systemic risk in G7 equity markets [16], underscoring the necessity of multi-factor collaborative modeling. To address the challenges of multi-scale feature extraction and fusion, various enhanced approaches have been proposed. Notably, attention-guided decomposition-reconstruction models [17] have been developed to strengthen the capability of capturing hierarchical temporal features. A recent systematic review [18] further highlights that deep learning applications in financial time-series forecasting are evolving from single-model paradigms toward integrated frameworks combining multiple techniques. To improve model adaptability to complex market dynamics, researchers have advanced a spectrum of methodologies—from cross-correlation analysis and cointegration tests to multivariate artificial intelligence approaches. The TIC-FusionNet framework [19], which integrates time-series decomposition with attention mechanisms, has notably advanced the collaborative modeling of multi-source temporal information. Nevertheless, existing methods still lack explicit modeling of interactions among variables. Consequently, predictions based solely on a single time series face skepticism regarding their reliability and generalizability, particularly in effectively defining the applicable boundaries for different prediction horizons.

A comprehensive review of existing research reveals four primary limitations: first, the selection of wavelet bases relies heavily on empirical experience and lacks a data-driven adaptive mechanism; second, the static allocation of attention weights struggles to adapt to abrupt shifts in market conditions; third, temporal encoding falls short in modeling non-integer periods; and finally, the high complexity of hybrid models hinders their real-time deployment. These issues constrain the performance of existing methods in practical financial forecasting, with notable shortcomings particularly evident in handling high-frequency data and responding to rapid market fluctuations.

To address these limitations, this study proposes a hybrid deep learning model, CBAM-LSTM-Time2Vec-KAN, based on optimized multi-module collaborative features, employing a systematic approach that integrates multiple technologies to tackle the aforementioned challenges. The core innovations of the proposed model are reflected in four dimensions:

- (1) At the wavelet transform level, we propose an adaptive wavelet basis selection algorithm based on information entropy. By comparing the decomposition performance of different wavelet bases on financial data, the algorithm automatically selects the most suitable wavelet basis function for a given dataset. In contrast to existing studies that employ a fixed wavelet basis [20], our adaptive selection mechanism better accommodates the distinct data characteristics under varying market conditions.
- (2) At the attention mechanism level, we introduce the Northern Goshawk Optimization (NGO) algorithm to guide the learning process of CBAM attention weights. Inspired by the intelligent hunting behavior of raptors, NGO offers a more efficient alternative to the complex risk modeling architectures in prior work [21,22]. By optimizing CBAM attention weights with NGO, our approach achieves comparable or superior risk control performance while significantly reducing model complexity, enabling attention weights to more accurately focus on critical features. Both theoretical analysis and empirical results confirm that the NGO algorithm guarantees global convergence in non-convex optimization problems.
- (3) At the temporal feature encoding level, we present a significant enhancement to Time2Vec by incorporating cosine harmonic terms to construct an augmented time encoder. This improved encoder not only preserves the original Time2Vec's ability to capture periodic patterns, but also effectively models more complex temporal structures—such as non-integer-period cycles and modulated periodic phenomena. Compared to studies relying on single temporal feature extraction methods [23,24] or multi-view feature fusion strategies [25], our framework synergistically combines wavelet-based multi-scale decomposition with Time2Vec's periodic encoding, thereby capturing the multi-temporal-scale characteristics of stock prices more comprehensively.
- (4) At the network architecture level, we replace the traditional multilayer perceptron (MLP) with a Kolmogorov–Arnold network (KAN) as the prediction head. Built upon the principle of function composition, KAN achieves more complex nonlinear mappings with far fewer parameters, maintaining high prediction accuracy while substantially reducing model complexity. In comparison with hybrid forecasting models in prior research [26], our KAN-based architecture strikes a better balance between model simplicity and predictive performance.
- (5) In terms of practical applications, the forecasting framework proposed in this study provides an effective tool for financial risk management. While reinforcement learning methods for portfolio optimization [27] have achieved significant success, our model leverages the combined advantages of wavelet denoising, Time2Vec temporal encoding, and KAN nonlinear fitting at the quantitative investment level. It can effectively filter out noise from stock data, uncover hidden temporal features and nonlinear fluctuation patterns, and accurately forecast stock price movements and volatility trends. This provides robust data support for individual investors and quantitative trading strategies, assisting investors in scientifically evaluating market conditions, optimizing trade timing, and rationally adjusting position structures, thereby enhancing the rationality of investment decisions and the stability of returns. At the financial risk management level, the model can precisely capture abrupt shifts in short-term volatility, inflection points in medium- to long-term trends, and potential market risk signals. This offers a quantitative basis for financial institutions, helping them proactively anticipate market risks, improve risk prevention and control systems, and enhance their capabilities in managing stock market price and liquidity risks.

2. Related Methods and Principles

2.1. Wavelet Transform (WT)

From the perspective of time–frequency analysis, any complex signal can be viewed as a superposition of components at different frequencies. The abrupt and discontinuous parts of a signal correspond to high-frequency components, while the smooth and stable parts belong to low-frequency components. Although traditional Fourier analysis can decompose a signal into its frequency spectrum using sinusoidal basis functions, it fails to provide information about how these frequency components evolve over time. To overcome this limitation, Morlet et al. [28] innovatively proposed the wavelet transform method. This approach employs wavelet basis functions with time–frequency localization properties to accurately capture a signal’s time–frequency characteristics.

The core of the wavelet transform lies in using a set of wavelet functions, scaled and translated in time, to approximate the signal. To ensure the mathematical completeness of the transform, these basis functions must satisfy two fundamental conditions: first, they must be integrable in both the time and frequency domains; second, their inner product with the signal must converge, and their mean value at zero frequency must be zero. The wavelet transform is defined as follows:

$$\begin{aligned} f(t, \alpha, \tau) &= [f(t), \psi_{\alpha, \tau}] \\ &= \frac{1}{|\alpha|^{\frac{1}{2}}} \int f(t) \psi\left(\frac{t-\tau}{\alpha}\right) dt \end{aligned} \quad (1)$$

In the equation: $\psi(t)$ is the continuous mother wavelet function; α is the scaling factor; τ is the translation factor. Both variables are continuous. In this case, the transform possesses infinitely many wavelet frequencies, allowing arbitrary scaling of the mother wavelet.

2.2. Northern Goshawk Optimization (NGO) Algorithm

The Northern Goshawk Optimization (NGO) algorithm is a novel intelligent optimization algorithm inspired by the unique hunting behavior of the northern goshawk. It achieves efficient optimization search by simulating two key phases of its hunting process. Compared to other optimization algorithms, the NGO algorithm effectively balances global and local search, significantly improving accuracy and stability. The core mechanism of the algorithm consists of two main phases:

(1) Exploration Phase (Global Search Phase)

This phase simulates the goshawk’s process of identifying and attacking prey. The algorithm performs an extensive global search during this stage. Specifically, it randomly selects potential solutions within the search space (simulating prey selection) and rapidly initiates attacks (evaluating and updating solutions), aiming to quickly locate potential optimal regions. Its mathematical expression is as follows:

$$P_i = X_k, i = 1, 2, \dots, N, k = 1, 2, \dots, i-1, i+1, \dots, N \quad (2)$$

$$x_{i,j}^{new,P_1} = \begin{cases} x_{i,j} + r(P_{i,j} - Ix_{i,j}), & F_{P_i} < F_i \\ x_{i,j} + r(x_{i,j} - P_{i,j}), & F_{P_i} \geq F_i \end{cases} \quad (3)$$

$$X_i = \begin{cases} X_i^{new,P_1}, & F_i^{new,P_1} < F_i \\ X_i, & F_i^{new,P_1} \geq F_i \end{cases} \quad (4)$$

In the equation: P_i represents the position of the target prey for the i northern goshawk; $P_{i,j}$ denotes the value of the prey’s position in the j dimension. X_k is the matrix composed of the vectors of the northern goshawks; $x_{i,j}^{new,P_1}$ denotes the j dimension of the i solution in

the first phase under the new state; F_i^{new,P_1} is the objective function value in the first phase; r represents a random number between 0 and 1; I takes a value of 1 or 2.

(2) Exploitation Phase (Local Optimization Phase):

This phase simulates the dynamic process of the goshawk chasing its prey and the prey's escape behavior. During this stage, the algorithm conducts a refined local search. By adjusting the search step size and direction, it gradually approaches the optimal solution, achieving precise optimization. The mathematical expression is as follows:

$$x_{i,j}^{new,P_2} = x_{i,j} + R(2r - 1)x_{i,j} \quad (5)$$

$$R = 0.02(1 - \frac{t}{T}) \quad (6)$$

$$X_i = \begin{cases} X_i^{new,P_2}, & F_i^{new,P_2} < F_i \\ X_i, & F_i^{new,P_2} \geq F_i \end{cases} \quad (7)$$

In the equation: R represents the attack range of the population; t is the current iteration number; T is the maximum number of iterations; F_i^{new,P_2} is the objective function value in the second phase.

2.3. Long Short-Term Memory Model

Long short-term memory (LSTM) is a special type of recurrent neural network (RNN) designed to address the vanishing gradient and exploding gradient problems that traditional RNNs face when processing long sequences. Its core idea involves introducing gating mechanisms and memory cells to capture long-range dependencies. The model consists of a forget gate, an input gate, and an output gate.

LSTM Forget Gate: Determines which information in the memory cell should be discarded. It uses a sigmoid activation function to produce an output between 0 and 1, controlling the extent to which information in the memory cell is retained or forgotten.

$$f_i^{(t)} = \sigma \left(b_i^f + \sum_j U_{i,j}^f x_j^{(t)} + \sum_j W_{i,j}^f h_j^{(t-1)} \right) \quad (8)$$

LSTM Input Gate: Controls the extent to which new information from the current time step updates the memory cell. It consists of a sigmoid layer and a tanh layer. The sigmoid layer regulates the opening and closing of the input gate, while the tanh layer processes the input information.

$$i_t = \sigma \left(b_i^g + \sum_j U_{i,j}^g x_j^{(t)} + \sum_j W_{i,j}^g h_j^{(t-1)} \right) \quad (9)$$

$$C_t = \tanh \left(b_i + \sum_j U_{i,j}^f x_j^{(t)} + \sum_j W_{i,j} h_j^{(t-1)} \right) \quad (10)$$

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \quad (11)$$

LSTM Output Gate: Determines which information from the memory cell will be output to the hidden state and influence the current time step's output. Similarly, it uses a sigmoid layer to generate values between 0 and 1, controlling the extent of information output.

$$h_i^{(t)} = \tanh(s_i^{(t)}) - o_i^{(t)} \quad (12)$$

2.4. Convolutional Block Attention Module

Convolutional block attention module (CBAM) is a lightweight attention mechanism module primarily used to enhance the feature extraction capability of convolutional neural networks. This module effectively filters key feature information and suppresses irrelevant feature interference through a dual attention mechanism (channel attention and spatial attention), thereby significantly improving model performance. Specifically, CBAM consists of two core sub-modules [29], as shown in Figures 1 and 2.

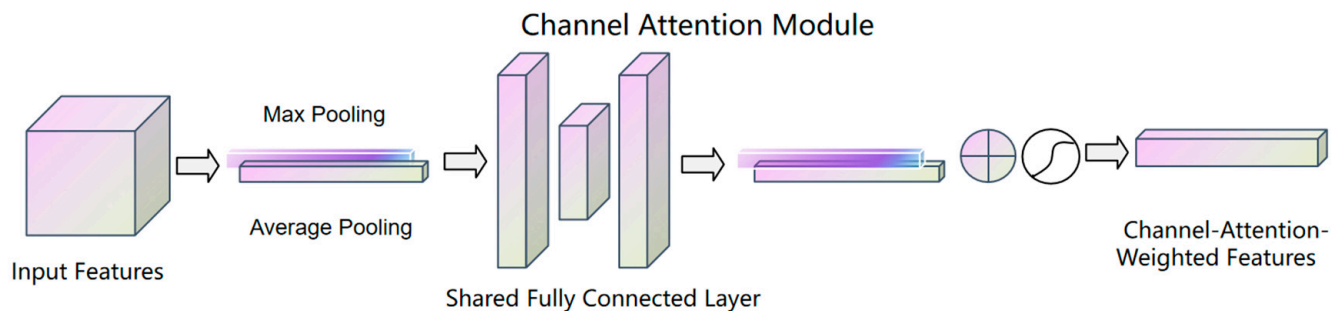


Figure 1. Channel Attention Module.

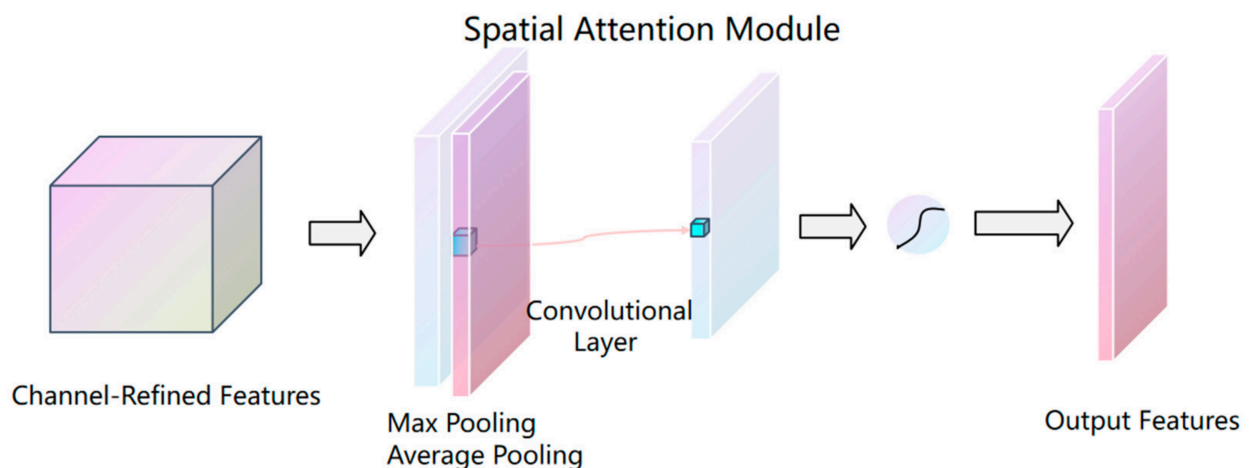


Figure 2. Spatial attention module.

CAM (channel attention module) automatically learns the weight distribution across different channels by analyzing the inter-channel relationships of feature maps. Spatial attention module (SAM) further analyzes the importance of features along the spatial dimensions based on the feature maps processed by the channel attention. By compressing the channel dimension and performing spatial attention calculations on the feature maps, it generates a spatial attention weight map.

CAM leverages inter-channel dependencies to determine which channels are more critical for the current task, thereby activating and generating the channel attention map.

$$\begin{aligned}
 M_c(F) &= \sigma \{ MLP \times [AvgPool(F)] + MLP \times [MaxPool(F)] \} \\
 &= \sigma \left\{ W_1 \left[W_0 \left(F_{avg}^c \right) \right] + W_1 \left[W_0 \left(F_{max}^c \right) \right] \right\}
 \end{aligned} \quad (13)$$

In the Equation(13): M_c represents the output channel attention features; MIP denotes the Multilayer Perceptron in the shared network; σ is the sigmoid function; MaxPool represents the max-pooled features; AvgPool represents the average-pooled features; AvgPool denotes average pooling; MaxPool denotes max pooling.

SAM can adaptively adjust the weights of spatial regions, enabling the network to focus more on spatial locations that contribute to the current task, thereby improving model performance. The calculation formula is as follows:

$$\begin{aligned} M_s(F_1) &= \sigma(f'^{\times 1} \{[AvgPool(F_1); MaxPool(F_1)]\}) \\ &= \sigma\left\{f^{7 \times 7} \left(\begin{bmatrix} F_{1avg}^s; F_{1max}^s \end{bmatrix} \right)\right\} \end{aligned} \quad (14)$$

In Equation (14): F_{1avg}^s represents the 2D feature map generated by average pooling, with dimensions $H \times W \times 1$; $F_{1avg}^s \in R^{1 \times H \times W}$, F_{1max}^s represents the 2D feature map generated by max pooling, with dimensions $H \times W \times 1$; $f^{7 \times 7}$ denotes the convolution operation with a filter size of 7×7 .

2.5. Time2Vec

Time2Vec is a temporal feature encoding technique that maps discrete time features into a continuous vector space by drawing inspiration from word embedding concepts, establishing a unified temporal representation method. This embedding layer can simultaneously capture periodic fluctuations and nonlinear trends in time series. Its unique vectorized representation not only overcomes the limitations of traditional temporal feature processing, but also supports cross-model feature sharing and information interaction. Characterized by its simple structure and strong adaptability, this module maintains stability against temporal scale variations and can be seamlessly integrated into various neural network architectures, providing an efficient embedded solution for time-series data analysis.

$$t2v(x)[i] = \begin{cases} \omega_{0h}x + \phi_{0h}, i = 0 \\ \sin(\omega_{ih}x + \phi_{ih}), 1 \leq i \leq k \end{cases} \quad (15)$$

In Equation (15): $t2v(x)[i]$ represents the i element; k is the embedding dimension; ω_{ih} and ϕ_{ih} are learnable parameters; the linear term $\omega_{0h}x + \phi_{0h}$ captures the non-periodic components of the time series; the periodic activation function $\sin(\omega_{ih}x + \phi_{ih})$ extracts the periodic components of the input data.

2.6. Kolmogorov–Arnold

The Kolmogorov–Arnold representation theorem is a seminal result in mathematics. The theorem states that any continuous multivariate function can be represented as a composition of a finite number of simpler functions—in formal terms, as nested univariate functions. In KANs, each node can be viewed as a flexible and learnable highly nonlinear mapping. By composing these mappings hierarchically, the network is able to effectively approximate complex functional relationships.

The advantage of KAN lies in its potential for efficient expression while maintaining a relatively compact model structure. KAN is built upon supervised learning tasks, with the goal of approximating a function f . In this supervised learning framework, the inputs x of all data points are mapped to their corresponding outputs y . By applying the Kolmogorov–Arnold theorem, this method can decompose any multivariate function into a series of univariate functions and summation operations, as illustrated in Figure 3.

Formula (16) is the theoretical Kolmogorov–Arnold representation theorem. Formula (17) extends this decomposition into a hierarchical deep network form, which is the forward propagation expression of a multi-layer KAN.

$$f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{p,q}(x_p) \right) \quad (16)$$

$$f(x) = \sum_{i_{L-1}=1}^{n_{L-1}} \phi_{L-1,i_{L-1}} \left(\sum_{i_{L-2}=1}^{n_{L-2}} \cdots \left(\sum_{i_2=1}^{n_2} \phi_{2,j_2,j_2} \left(\sum_{i_1=1}^{n_1} \phi_{1,j_1,j_1} \left(\sum_{i_0=1}^{n_0} \phi_{0,j_0,j_0}(x_{i_0}) \right) \right) \right) \cdots \right) \quad (17)$$

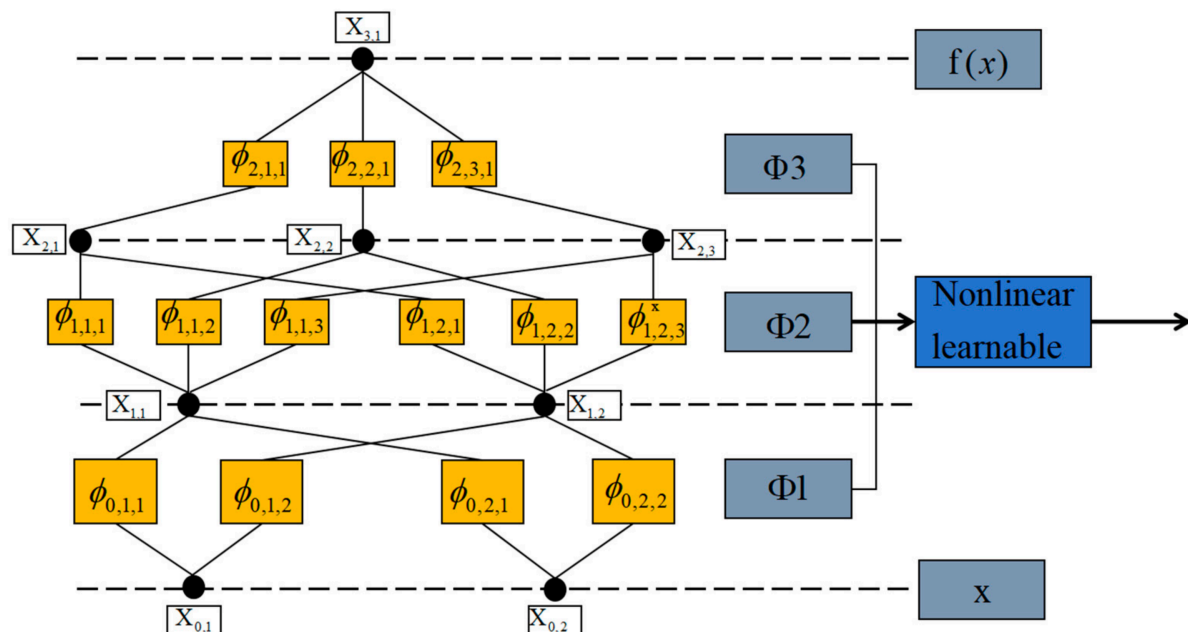


Figure 3. Structure diagram of the KAN.

2.7. Construction of the Combined Model

(1) Data Preprocessing and Multimodal Feature Extraction

First, preprocessing is performed on the original stock price data, including time column parsing, missing value handling, and numerical type conversion. Wavelet transform (WT) technology is employed to decompose the target stock price sequence into multiple frequency components at various scales. A reconstructed signal is generated through wavelet reconstruction, which, along with the original closing price, serves as input features for the subsequent model. Additionally, temporal features (day of the week, day of the month, month) are extracted and encoded using Time2Vec to generate embeddings with periodic characteristics, forming a multimodal feature input system.

(2) Temporal Feature Extraction and Continuous-Time Modeling

The LSTM module acts as the core temporal feature extractor, receiving dual-channel inputs of the wavelet-processed closing price and the reconstructed signal. A two-layer stacked LSTM structure captures long-term dependencies in the time series, outputting hidden states rich in temporal information. This better describes the continuous change characteristics of stock prices.

(3) Attention Mechanism Optimization and Feature Fusion

The CBAM attention module consists of two sub-modules: channel attention and spatial attention. It evaluates the importance of each feature channel through a squeeze-and-excitation mechanism and focuses on key time steps. The dual attention mechanisms work synergistically to enhance the weights of important features and suppress noise interference. The processed features are concatenated with the Time2Vec-generated temporal encodings, forming a comprehensive feature representation that integrates both temporal information and temporal semantics, providing a rich feature foundation for the final prediction.

(4) Prediction Output and Parameter Optimization

The KAN prediction head receives the concatenated comprehensive features and performs nonlinear transformation and feature fusion through fully connected layers, ultimately outputting the prediction results. The Northern Goshawk Optimization (NGO) algorithm is used to collaboratively optimize key hyperparameters in the model, including the KAN hidden layer dimension and CBAM compression ratio. A multi-objective fitness function evaluates the performance of parameter combinations, and an elite retention strategy combined with a Lévy flight mechanism is employed for iterative optimization, ultimately determining the globally optimal parameter set. The detailed flowchart is illustrated in Figure 4.

The overall prediction procedure of the proposed WT-NGO-LSTM-CBAM-Time2Vec-KAN model is summarized in Algorithm 1.

Algorithm 1. Prediction Procedure of WT-NGO-LSTM-CBAM-Time2Vec-KAN

Input: Stock price series $P = \{p_1, p_2, \dots, p_T\}$, lookback window L , prediction horizon H

Output: Predicted stock price $\hat{p}_{T+H}$

1: Step 1: Multi-scale Feature Decomposition (WT)

2: Apply Wavelet Transform to decompose P into approximation coefficient C_A and detail coefficients C_D^k .

3: $[C_A, C_D^k] = \text{WT}(P)$

4: Concatenate multi-scale features:

5: $X_{wt} = [C_A, C_D^1, C_D^2, \dots, C_D^k]$

6: Step 2: Temporal Dependency Modeling (LSTM)

7: Feed X_{wt} into LSTM network:

8: $h_t = \text{LSTM}(X_{wt}, \Theta_{LSTM})$

9: Step 3: Attention Refinement (CBAM)

10: Channel attention weighting:

11: $h_t' = \text{CBAM}_{\text{channel}}(h_t)$

12: Spatial attention refinement:

13: $h_t'' = \text{CBAM}_{\text{spatial}}(h_t')$

14: Step 4: Time Encoding (Time2Vec)

15: Encode calendar features:

16: $t_{enc} = \text{Time2Vec}(\text{dow}, \text{dom}, \text{month})$

17: Step 5: NGO-inspired Adaptive Fusion

18: Exploration feature:

19: $f_{exp} = \sigma(\text{MLP}(h_t''))$

20: Exploitation feature:

21: $f_{ext} = \sigma(\text{Conv1D}(t_{enc}))$

22: Adaptive fusion:

23: $f_{fus} = \alpha \cdot f_{exp} + \beta \cdot f_{ext}$

24: Step 6: Nonlinear Regression (KAN)

25: Final prediction:

26: $\hat{p}_{T+H} = \text{KAN}(f_{fus}, \Theta_{KAN})$

27: return $\hat{p}_{T+H}$

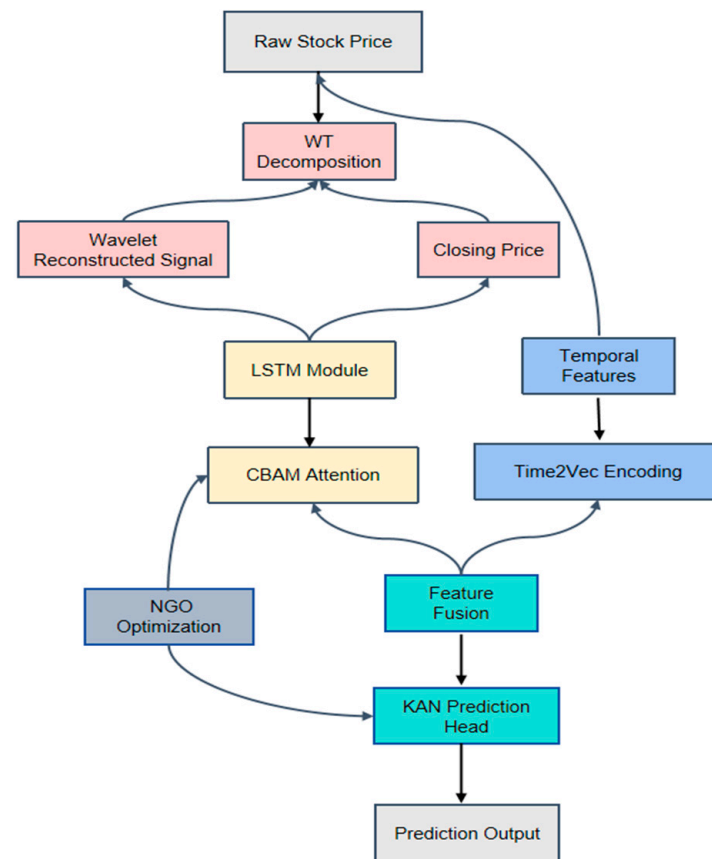


Figure 4. Flowchart of the model structure and pseudocode.

3. Experimental Analysis

3.1. Data Preparation

The experimental data in this study were sourced from Tonghuashun. To validate the effectiveness of the proposed model, three representative stocks were selected for empirical analysis based on their time-series characteristics and volatility in risk-return profiles—specifically, one stock from each of the financial, infrastructure, and consumer manufacturing sectors: Agricultural Bank of China (Stock Code: 601288), Yantian Port (Stock Code: 000088), and Gree Electric Appliances (Stock Code: 000651). These are hereafter referred to as Q1, Q2, and Q3, respectively. Historical data for these stocks spanning from 15 July 2010 to 5 March 2025 were collected and used as the experimental dataset. Prior to modeling, the dataset was split into training and testing sets at a ratio of 8:2. Comparative experiments with different models and ablation studies were conducted during the prediction phase.

3.2. Evaluation Metrics

3.2.1. Forecasting Evaluation Metrics

To assess the prediction accuracy of the WT-NGO-LSTM-CBAM-Time2Vec-KAN model, the mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) were employed as evaluation metrics. Their formulas are as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (18)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (19)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (20)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (21)$$

For MAE, RMSE, and MAPE, smaller values indicate smaller errors and better prediction performance of the model. For R^2 , a value closer to 1 indicates a better fit of the model.

3.2.2. Backtesting Evaluation Metrics

To verify the practical performance of model-based trading strategies, the Sharpe ratio is adopted to measure the performance of each model during backtesting. The calculation formula is given as follows.

$$\text{Sharpe Ratio} = \frac{r_p - r_f}{\sigma_p} \quad (22)$$

$$r_p = \left((1 + P)^{\frac{250}{n}} - 1 \right) \times 100\% \quad (23)$$

For the Sharpe ratio in backtesting evaluation metrics, a larger value signifies higher excess returns per unit risk and stronger practical profitability of the model-driven trading strategy.

3.3. Experimental Setup

3.3.1. WT Feature Extraction Module

The WT feature extraction module is based on wavelet transform technology. By decomposing the stock price series at multiple scales, it effectively separates high-frequency noise from low-frequency trend components. This module breaks down the original signal into subsequences of different frequency bands, extracting feature components that represent short-term fluctuations, medium-term cycles, and long-term trends. It provides a multi-scale feature representation with enhanced noise robustness for subsequent prediction models. The wavelet decomposition diagrams for each model are shown in Figures 5–7.

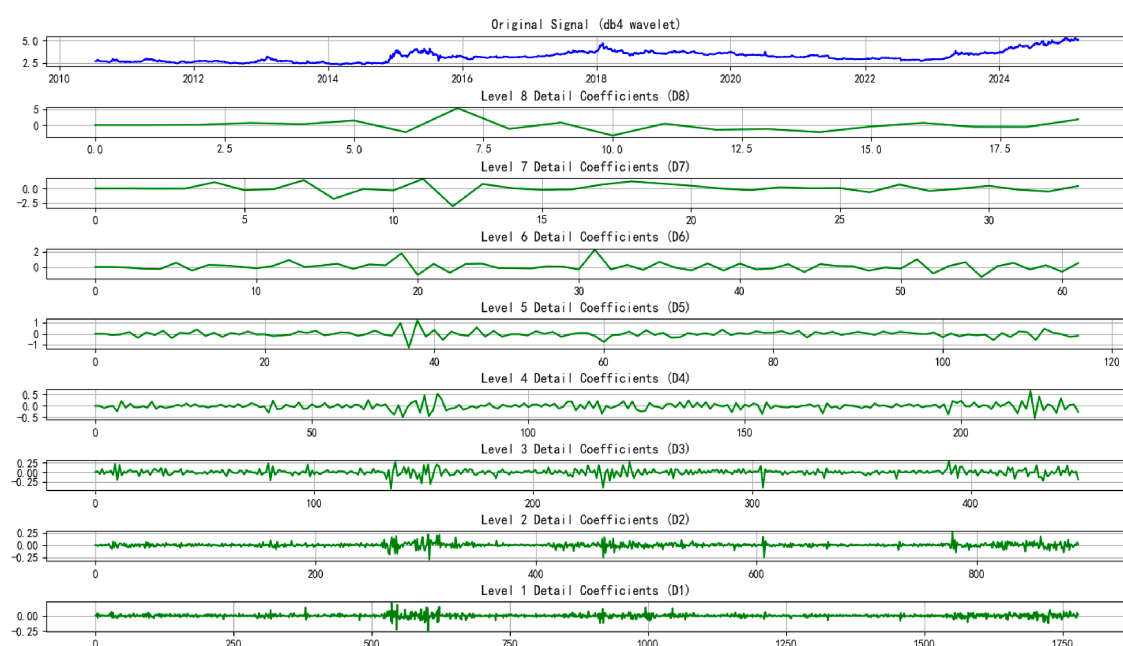


Figure 5. WT decomposition results for Q1.

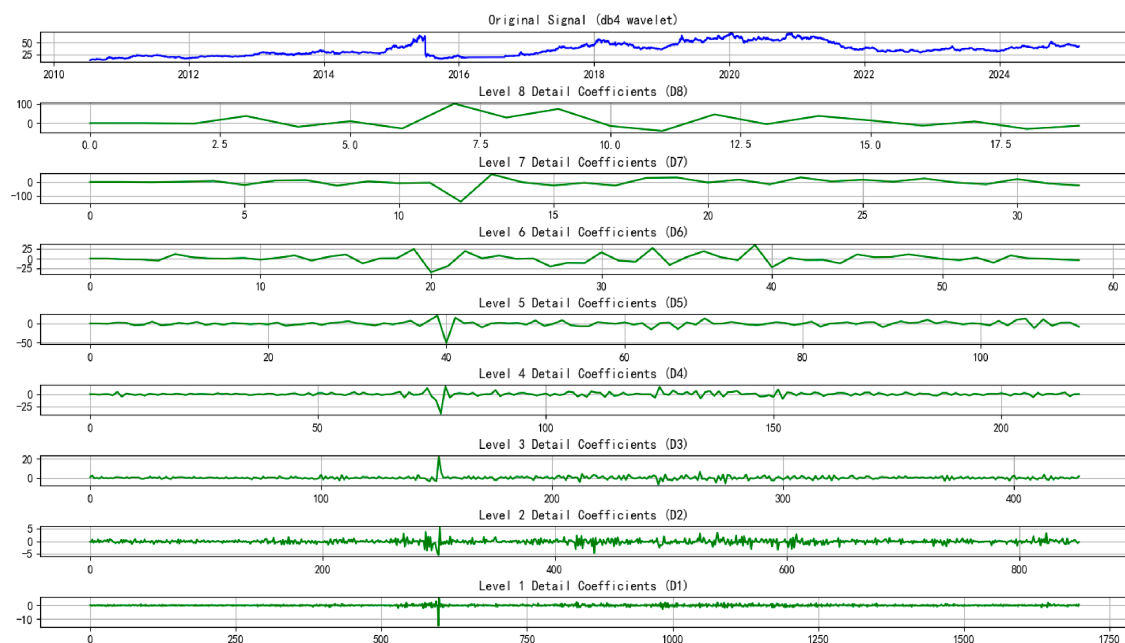


Figure 6. WT decomposition results for Q2.

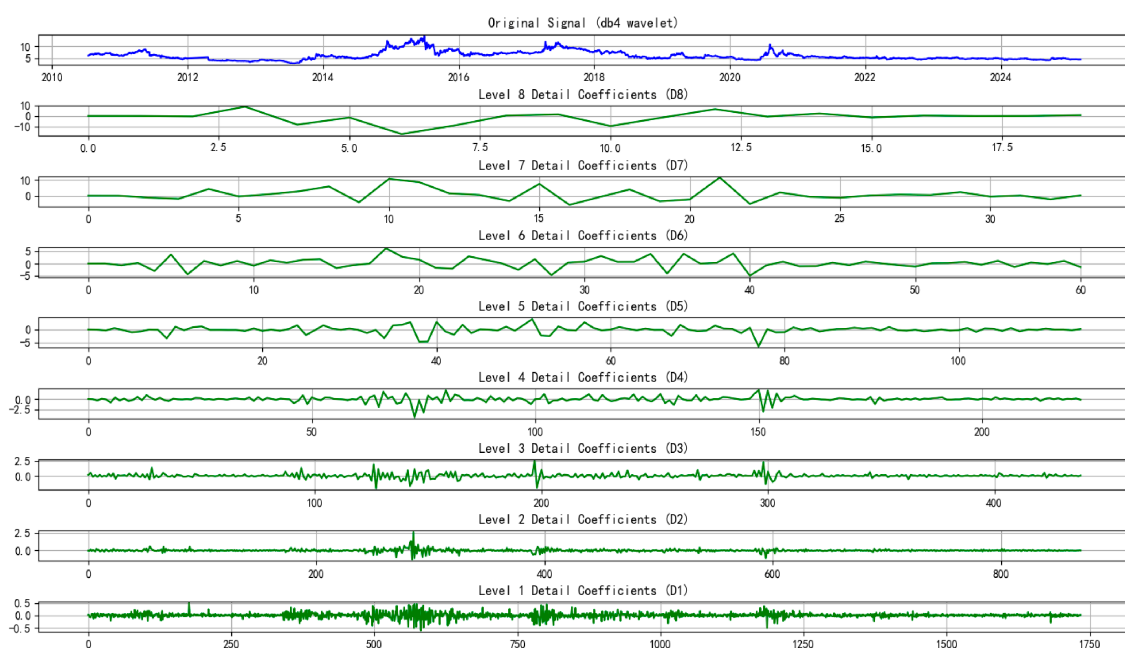


Figure 7. WT decomposition results for Q3.

3.3.2. Parameters of the Prediction Model

For the LSTM-CBAM-Time2Vec-KAN prediction model, an automated and robust search for model architecture and training configuration was conducted to obtain the optimal parameters. The specific parameter settings are shown in Table 1:

Table 1. Parameter settings of the prediction model.

	Parameter	Value/Range
LSTM	input_dim	2
	hidden_dim	64
	output_dim	2

Table 1. Cont.

	Parameter	Value/Range
CBAM	channels	64
	Reduction_ratio	4
	activation	relu
Time2Vec	output_dim	8
	activation	sin
KAN	input_dim	80
	hidden_dim	64
	output_dim	1
	activation	silu

3.4. Comparative Analysis Experiments

3.4.1. Comparative Experiments of Different Models

To validate the predictive performance of the WT-NGO-LSTM-CBAM-Time2Vec-KAN model, comparative experiments were conducted with models using different approaches, including CNN-Transformer, GA-LSTM, PCA-Transformer, and GA-CNN-LSTM. The results are shown in Figures 8–13.

As illustrated in the prediction results from the multi-model comparison experiments, the proposed LSTM-CBAM-Time2Vec-KAN model achieved the highest fidelity between its predicted trajectory and the actual stock prices compared to conventional models such as CNN-Transformer and GA-LSTM, significantly reducing prediction errors—particularly at points of sharp market volatility. The corresponding training loss curves further show that our model not only converged faster than all baselines, but also exhibited a remarkably stable training process, without the severe loss oscillations commonly observed in other models during mid-training iterations. This underscores its superior optimization characteristics and training stability. These experimental results demonstrate that the proposed LSTM-CBAM-Time2Vec-KAN model achieved the smallest prediction error during periods of high volatility, providing strong evidence for the effectiveness of the multi-module collaborative design in enhancing forecasting accuracy.

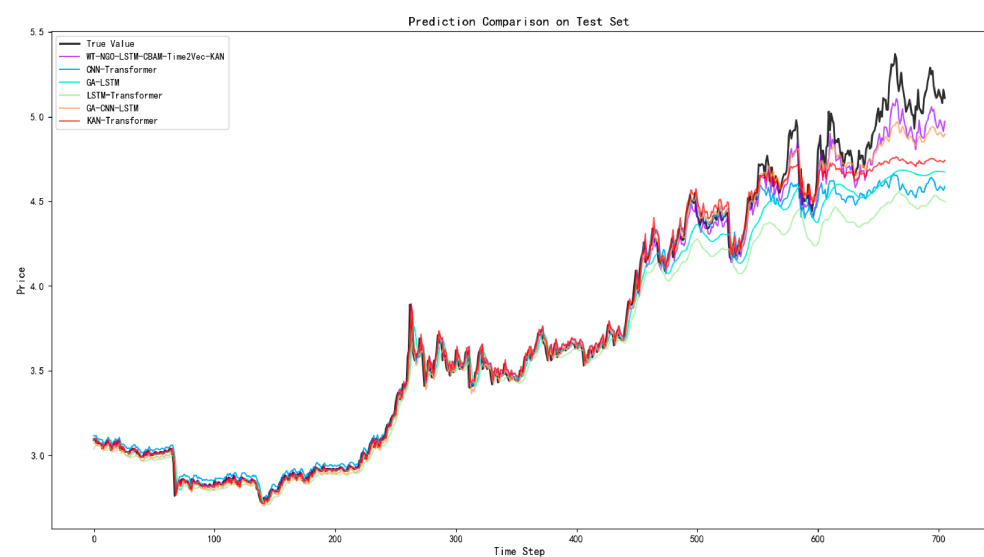


Figure 8. Comparative experimental results of single models for Q1.

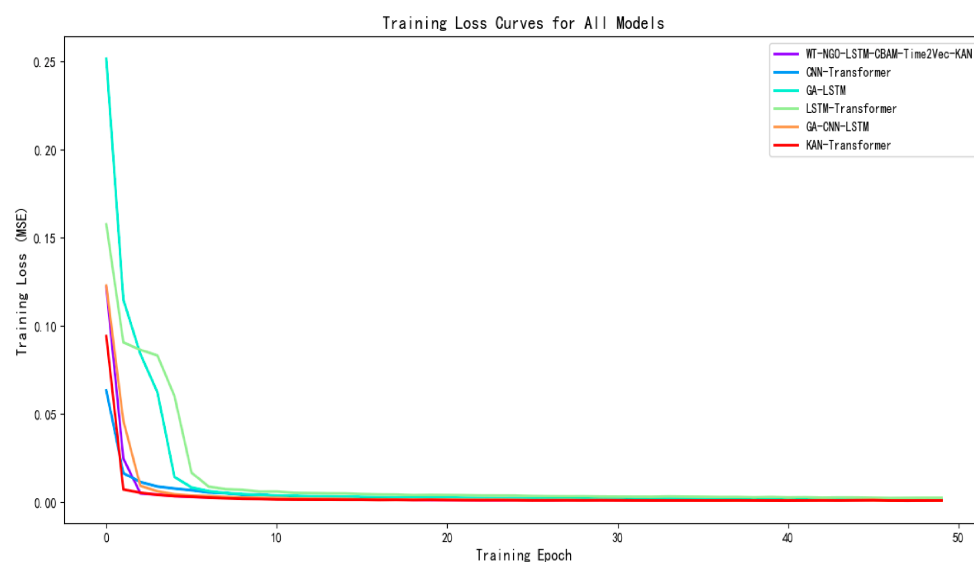


Figure 9. Comparative loss diagram of single models for Q1.

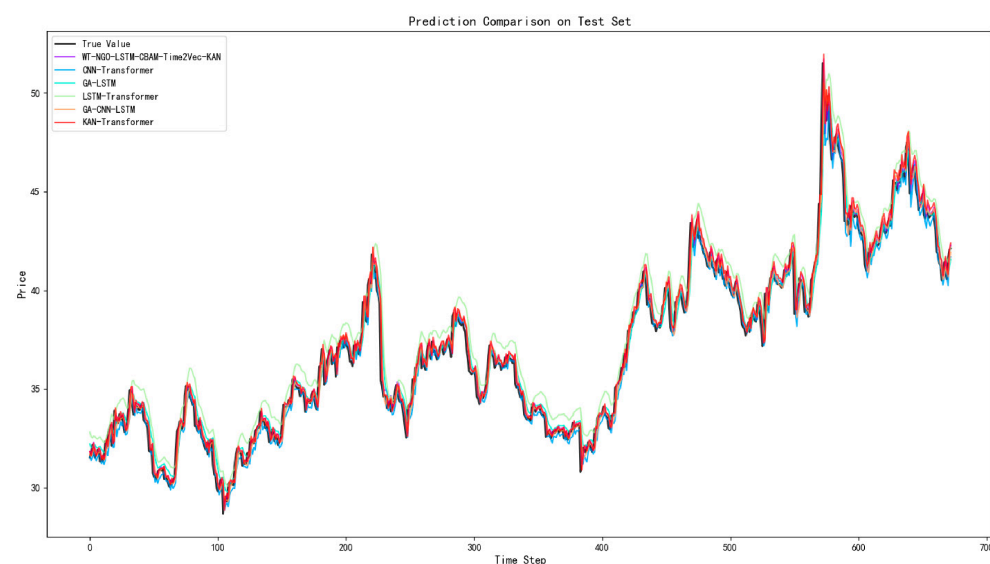


Figure 10. Comparative experimental results of single models for Q2.

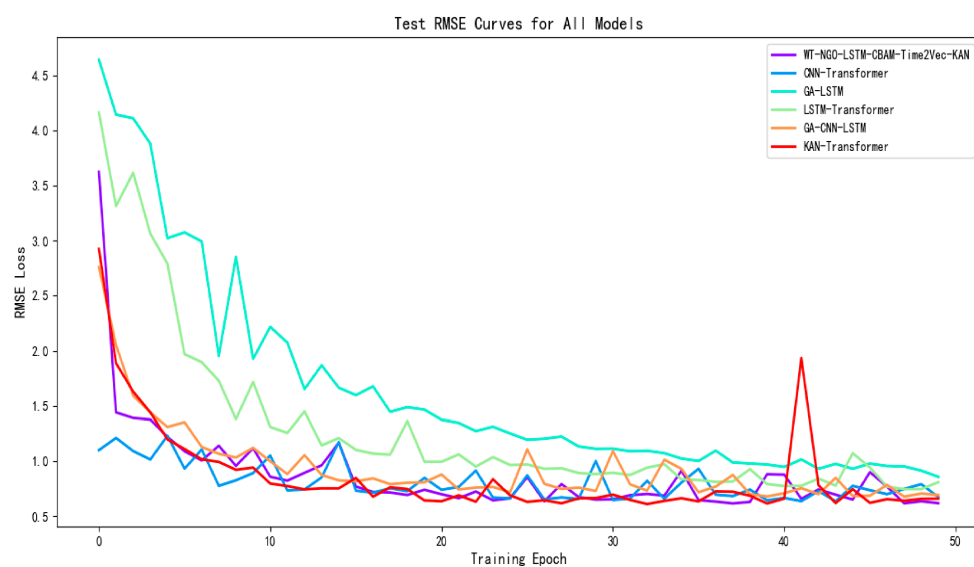


Figure 11. Comparative loss diagram of single models for Q2.

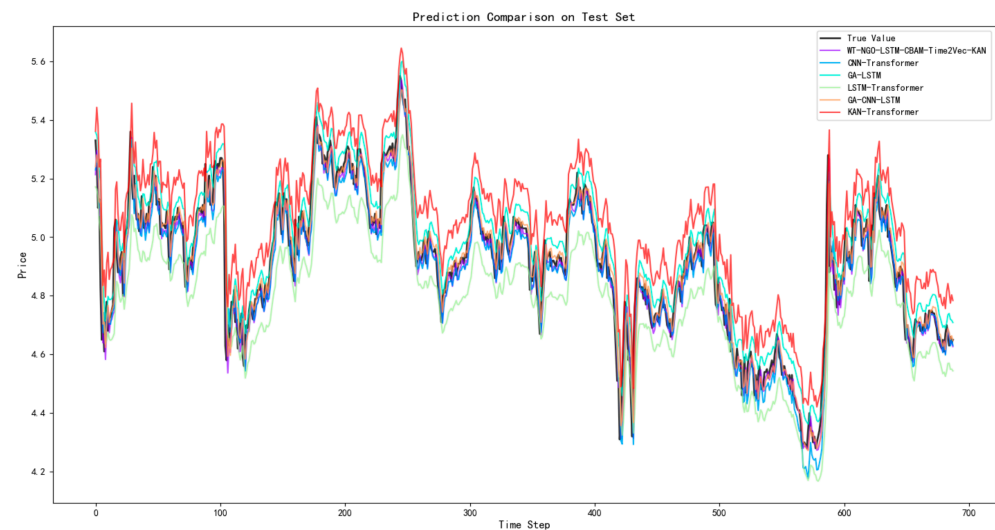


Figure 12. Comparative experimental results of single models for Q3.

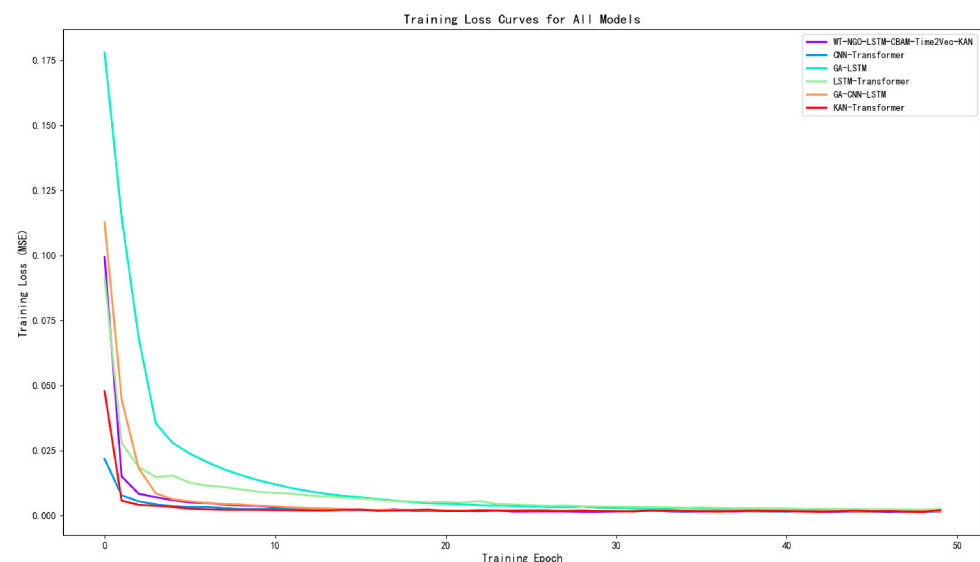


Figure 13. Comparative loss diagram of single models for Q3.

3.4.2. Ablation Experiments

To verify the contribution of each key component in the proposed model to predictive performance, this study designed multiple ablation experiments and conducted comparative analyses under the same dataset and evaluation framework. The experimental design encompasses all core modules of the model, including the wavelet transform (WT) pre-processing module, the Northern Goshawk Optimization (NGO) algorithm, the CBAM attention mechanism, the Time2Vec temporal encoder, and the Kolmogorov–Arnold network (KAN) prediction head. Specific modules were systematically removed or replaced to comprehensively evaluate their individual and synergistic roles in feature extraction, parameter optimization, temporal attention enhancement, periodic information embedding, and multimodal fusion. The results are presented in Figures 14–19.

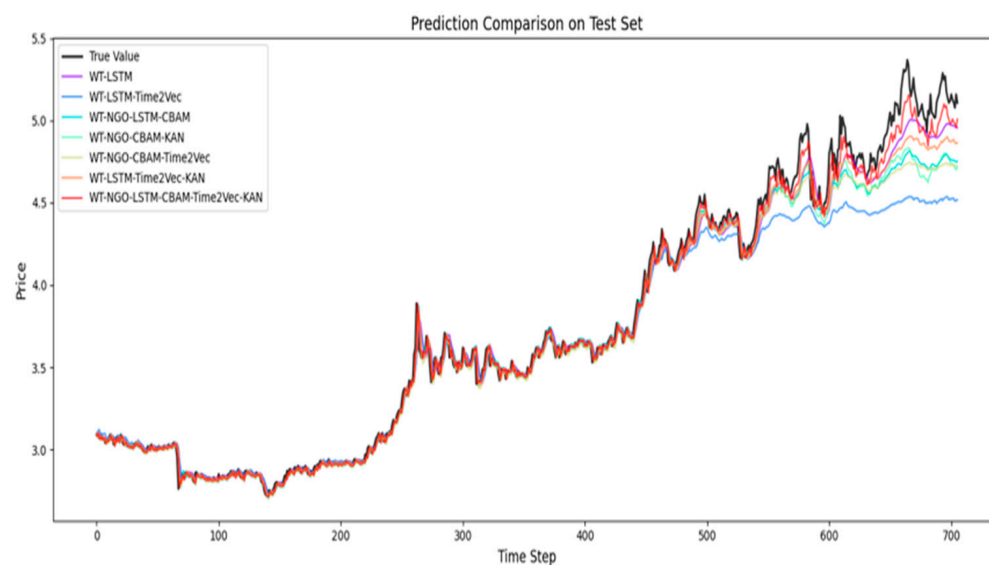


Figure 14. Different model comparison results for Q1.

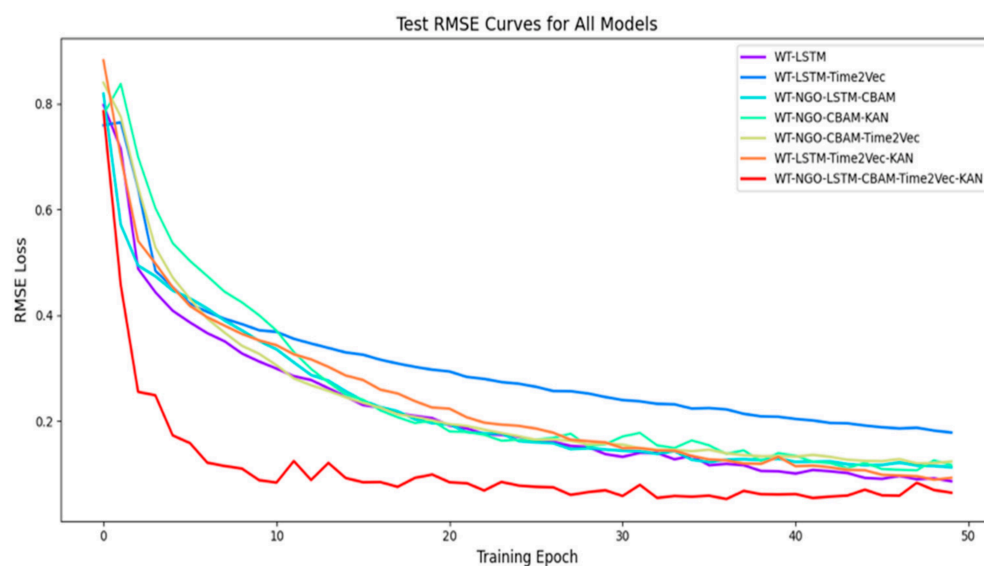


Figure 15. Comparative loss diagram of different models for Q1.

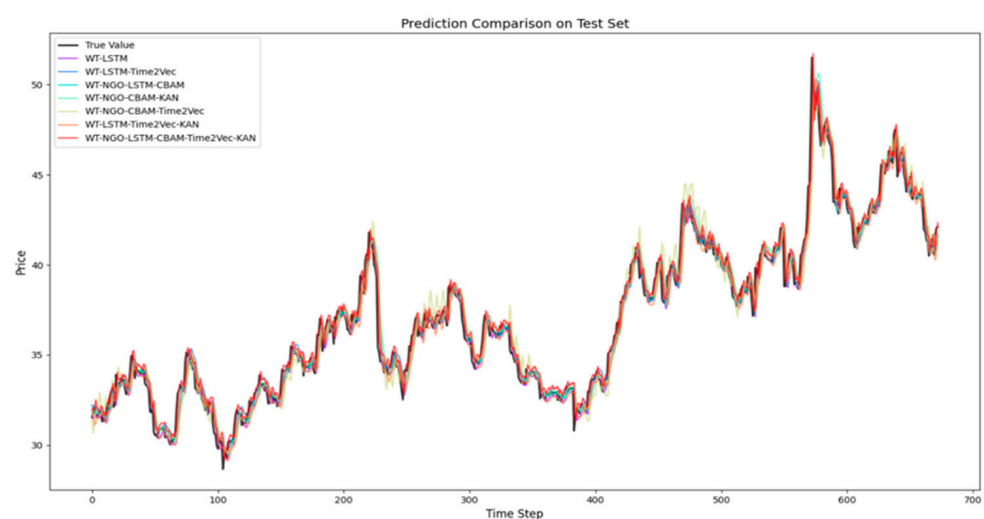


Figure 16. Different model comparison results for Q2.

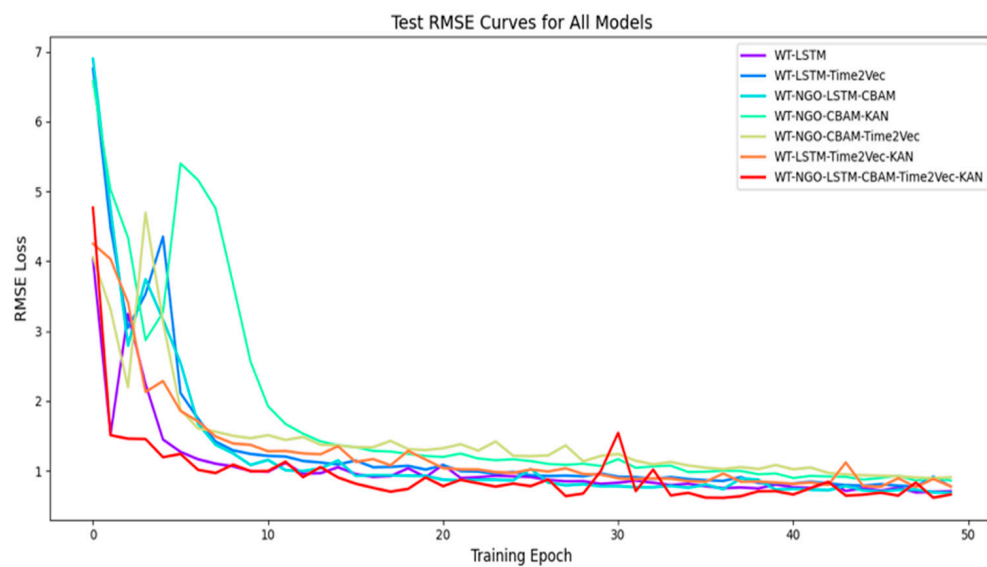


Figure 17. Comparative loss diagram of different models for Q2.

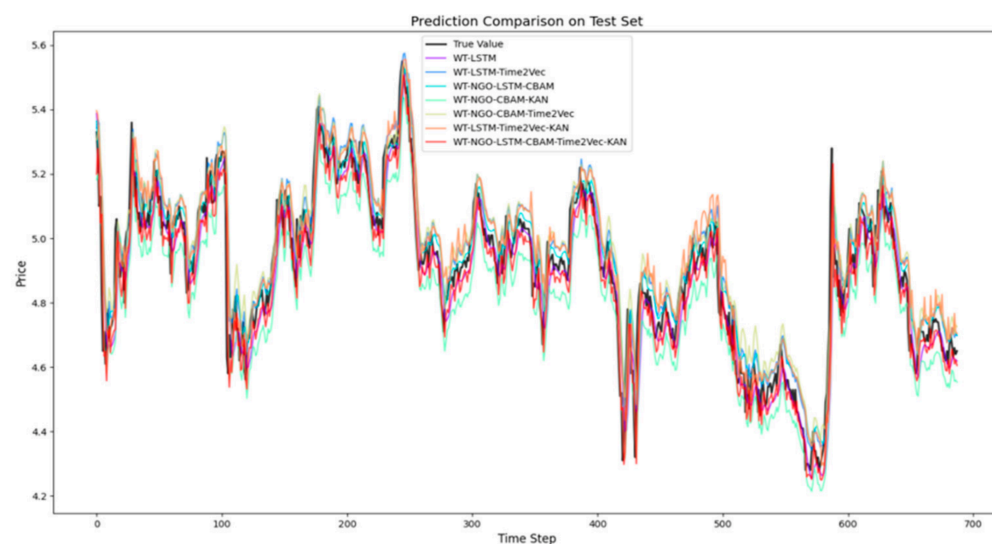


Figure 18. Different model comparison results for Q3.

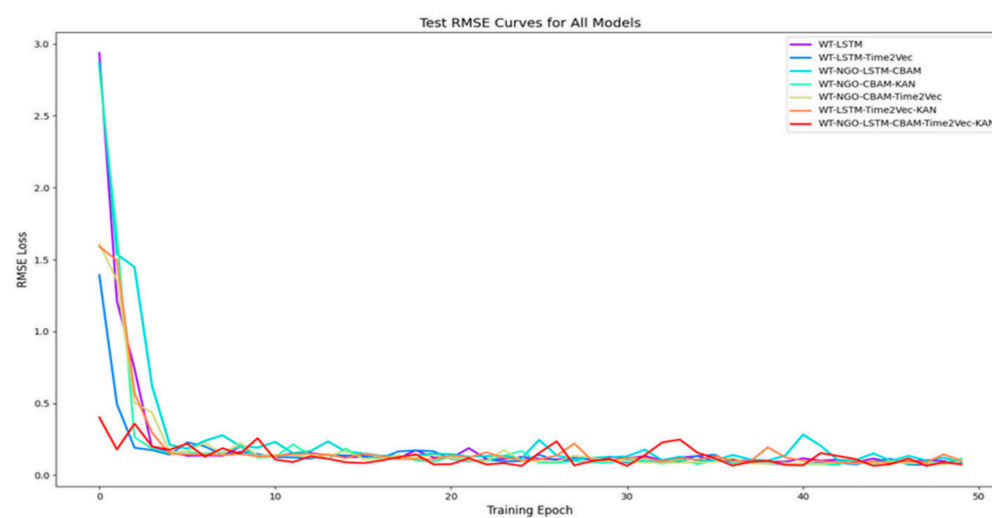


Figure 19. Comparative loss diagram of different models for Q3.

As shown in the ablation study's prediction performance comparison plots, the model's ability to fit stock price fluctuations improved significantly as each proposed module was incrementally incorporated. The full model achieved the smallest deviation from the ground truth at critical turning points, with its prediction curve closely aligning with the actual values. Training loss curves further revealed that our model not only converged faster than baseline models, but also stabilized at a consistently lower loss level, demonstrating its overall advantages in both training efficiency and stability. Moreover, the ablation experiments clearly indicate that removing any single component leads to noticeable prediction lag around peaks and troughs. In particular, omitting the Time2Vec encoding substantially degraded the model's capacity to capture periodic patterns. These results strongly confirm the irreplaceable role of each module in temporal feature extraction and validate the rationality and effectiveness of the overall model architecture.

3.5. Prediction Error Analysis of Each Model

This study selected RMSE, MAE, MAPE, and R^2 as the evaluation metrics for model prediction performance. These metrics offer complementary strengths: RMSE places a heavier penalty on large prediction errors; MAE provides a balanced measure of the overall average error; MAPE eliminates the influence of stock price magnitudes to enable cross-sectional comparisons; and R^2 quantifies the model's explanatory power in fitting temporal fluctuations. Together, this combination of metrics ensures a comprehensive and objective evaluation of time-series forecasting performance. Using the aforementioned evaluation formulas, we calculated and assessed the prediction results for the three stocks, Q1, Q2, and Q3, to obtain the corresponding metric values. Tables 2 and 3 present the comparative evaluation metrics across different models and the results of the ablation study, respectively.

Table 2. Comparison of evaluation metrics for the different models.

	Model	RMSE	MAE	MAPE/%	R^2 /%
Q1	GA-CNN-LSTM	0.1443	0.0823	1.8414	96.49
	LSTM-Transformer	0.0898	0.0627	1.5273	98.64
	GA-LSTM	0.1038	0.0635	1.4701	98.18
	CNN-Transformer	0.1812	0.0992	2.2253	94.46
	KAN-Transformer	0.1316	0.0711	1.6089	97.08
	WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.0616	0.0470	1.2384	99.36
Q2	GA-CNN-LSTM	0.9030	0.6503	1.7007	95.85
	LSTM-Transformer	1.2710	1.0599	2.8898	92.78
	GA-LSTM	0.9275	0.6904	1.8430	95.62
	CNN-Transformer	0.8403	0.6530	1.7688	96.41
	KAN-Transformer	0.6778	0.4827	1.2779	97.66
	WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.6409	0.4270	1.1357	98.41
Q3	GA-CNN-LSTM	0.0814	0.0604	1.2440	94.79
	LSTM-Transformer	0.0832	0.0593	1.2180	95.30
	GA-LSTM	0.0917	0.0630	1.2877	92.77
	CNN-Transformer	0.0819	0.0597	1.2130	95.66
	KAN-Transformer	0.0733	0.0528	1.0882	90.91
	WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.0730	0.0522	1.0676	97.98

Table 3. Comparison of evaluation metrics for the ablation experiments.

	Model	RMSE	MAE	MAPE/%	R ² /%
Q1	WT-LSTM	0.0986	0.0629	1.4840	98.36
	WT-LSTM-Time2Vec	0.2221	0.1254	2.7513	91.68
	WT-NGO-LSTM-CBAM	0.1351	0.0765	1.7218	96.92
	WT-NGO-CBAM-KAN	0.1347	0.0793	1.8112	96.94
	WT-NGO-CBAM-Time2Vec	0.1468	0.0859	1.9480	96.37
	WT-LSTM-Time2Vec-KAN	0.1065	0.0640	1.4737	98.09
	WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.0680	0.0458	1.1097	99.22
Q2	WT-LSTM	0.7361	0.5169	1.3822	97.04
	WT-LSTM-Time2Vec	0.8034	0.5851	1.5769	96.71
	WT-NGO-LSTM-CBAM	0.7155	0.4985	1.3319	97.93
	WT-NGO-CBAM-KAN	0.8904	0.6419	1.7176	95.96
	WT-NGO-CBAM-Time2Vec	0.9306	0.6821	1.8309	95.59
	WT-LSTM-Time2Vec-KAN	0.8125	0.5839	1.5647	96.64
	WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.6832	0.5006	1.3472	98.62
Q3	WT-LSTM	0.0944	0.0665	1.3542	94.93
	WT-LSTM-Time2Vec	0.0930	0.0716	1.4704	95.38
	WT-NGO-LSTM-CBAM	0.0939	0.0675	1.3922	95.08
	WT-NGO-CBAM-KAN	0.1187	0.1018	2.0662	86.15
	WT-NGO-CBAM-Time2Vec	0.1008	0.0796	1.6451	92.83
	WT-LSTM-Time2Vec-KAN	0.1024	0.0787	1.6154	92.27
	WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.0783	0.0589	1.1977	98.64

As evidenced by the above experimental results, comprehensive evaluation across multiple metrics—including MAE, RMSE, MAPE, and R²—consistently demonstrated that the complete model significantly outperformed both alternative models and all ablation variants across all evaluation criteria. This indicates that each module contributes irreplaceable gains to the overall predictive performance. For instance, ablation experiments replacing the KAN resulted in weakened multi-feature collaboration and degraded prediction accuracy. These findings collectively confirm that the proposed multi-module collaborative architecture effectively enhances the overall performance and robustness of stock price prediction through the synergistic integration of its components, further validating the rationality and necessity of the model design.

3.6. Analysis of Stock Price Prediction Performance Across Different Forecast Horizons

Based on the experiments conducted on the three stocks in the previous section, the stability and superior performance of the proposed model in stock price prediction were validated. In those experiments, the target variable was always the next-day closing price. To further evaluate the model's predictive effectiveness across different forecast horizons, we conducted additional experiments by setting the closing prices at the one-day, three-day, and seven-day horizons as the prediction targets, respectively. The resulting evaluation metrics for each forecast horizon are presented in Table 4.

From the evaluation metrics, it can be observed that as the forecast horizon increases, the model's predictive performance gradually declines. The fitting accuracy decreased from predicting the next-day closing price to the seven-day closing price. However, although the prediction accuracy for the three-day closing price was somewhat lower than that for the next-day closing price, it still maintained a relatively high level of fit and remains effective for stock price prediction.

In summary, the proposed prediction model in this study achieves strong performance in forecasting both next-day and three-day closing prices. While its performance is only

moderate for the longer seven-day horizon, the model is particularly well-suited for short-to medium-term stock price prediction and delivers superior predictive results within this range.

Table 4. Results of evaluation indicators for stock price prediction under different date spans.

Different Forecast Horizons		RMSE	MAE	MAPE/%	R ² /%
Q1	Next day's closing price	0.1391	0.1038	1.0440	98.62
	3-day closing price	0.1754	0.1157	1.1147	97.81
	7-day closing price	0.2641	0.1781	1.7204	95.04
Q2	Next day's closing price	0.1637	0.1148	1.0153	98.09
	3-day closing price	0.1750	0.2380	1.1155	96.94
	7-day closing price	0.1920	0.2392	1.2287	94.67
Q3	Next day's closing price	0.1222	0.0883	0.8828	98.94
	3-day closing price	0.1721	0.1489	1.4157	96.49
	7-day closing price	0.2281	0.1647	1.6468	95.10

3.7. Analysis of Simulated Trading Performance Under Different Trading Strategies

In Sections 3.5 and 3.6, the excellent and stable prediction performance of the constructed model on stock price series and its stability in medium-and short-term stock price forecasting were verified respectively. However, high prediction accuracy of a model in stock price forecasting does not necessarily translate to favorable returns or risk mitigation in real investment transactions. Therefore, simulated trading is conducted in this section to validate the practical performance of strategies based on the proposed models.

The model-driven trading strategy adopted in this experiment operates as follows: open a position at the buy signal and hold the stock until the model predicts an upcoming price decline, then close the position. The Sharpe ratio is selected as the evaluation metric for trading strategies, where a higher Sharpe ratio indicates greater excess returns per unit risk.

Simulated trading experiments were carried out on the test sets of three individual stocks, and the results are summarized in Table 5. As can be observed from the table, the strategy constructed based on the proposed model achieved a higher Sharpe ratio compared with all other model-based strategies. This verifies that the proposed model can generate high and stable returns when applied as an investment strategy in stock market trading.

Table 5. Simulated trading results of various trading strategies.

Model	Sharpe Ratio		
	Q1	Q2	Q3
LSTM-Transformer	0.33	0.23	0.27
GA-LSTM	0.26	0.40	0.38
GA-CNN-LSTM	0.34	0.47	0.41
CNN-Transformer	0.74	0.61	0.65
KAN-Transformer	0.77	0.73	0.74
WT-LSTM	0.55	0.41	0.47
WT-LSTM-Time2Vec	0.62	0.71	0.67
WT-NGO-LSTM-CBAM	0.30	0.46	0.35
WT-NGO-CBAM-KAN	0.48	0.55	0.51
WT-NGO-CBAM-Time2Vec	0.37	0.28	0.30
WT-LSTM-Time2Vec-KAN	0.46	0.54	0.45
WT-NGO-LSTM-CBAM-Time2Vec-KAN	0.84	0.83	0.87

4. Conclusions

To address challenges such as the strong nonlinearity, high non-stationarity, feature redundancy, and limited modeling capacity inherent in stock data, this paper proposes a stock prediction model named LSTM-CBAM-Time2Vec-KAN. The following conclusions are drawn based on experimental validation and error analysis:

(1) Validation of module effectiveness and demonstration of synergistic advantages

The core innovation of the proposed WT-NGO-LSTM-CBAM-Time2Vec-KAN model lies in its modular architecture. Experiments confirm that the wavelet transform (WT) preprocessing effectively extracts multi-resolution features, significantly improving the signal-to-noise ratio of the raw data. The NGO module enhances the modeling of complex stock price dynamics through continuous-time modeling, effectively complementing the LSTM in temporal feature extraction. The CBAM attention mechanism precisely focuses on key features and time steps through adaptive weighting across channel and spatial dimensions, thereby suppressing information redundancy. Meanwhile, Time2Vec encoding transforms discrete temporal information into embeddings with periodic patterns, greatly enriching the contextual information for the model. Finally, the prediction head constructed with KAN demonstrates superior nonlinear fitting capabilities compared to traditional MLPs. Rather than a simple stacking of components, these modules form a synergistically enhanced organic whole through end-to-end training.

(2) Significant improvement in prediction accuracy and robustness

In comparative experiments across multiple stock datasets, the proposed model significantly outperformed baseline models including LSTM, GRU and Transformer in key evaluation metrics such as RMSE, MAE and MAPE, which demonstrates its outstanding fitting capability and generalization ability. More importantly, error analysis revealed that the prediction curve of the proposed model deviated less from the ground truth at critical moments featuring sharp market price swings or trend reversals, exhibiting stronger robustness. This indicates that the proposed framework possesses superior capacity to capture and adapt to the non-stationary and nonlinear characteristics of stock data. Furthermore, prediction experiments over different time horizons verified the practicality and precision of the constructed model for medium-and short-term stock price forecasting. The above results once again confirm the powerful ability of the proposed architecture to capture and adapt to the non-stationarity and nonlinearity of stock data, and validate the rationality of the model structural design as well as the synergistic effectiveness of its multi-module integrated features. Finally, simulated trading experiments were conducted on the test sets of three individual stocks. The results show that the investment strategy built upon the proposed model achieved a higher Sharpe ratio than all other model-based strategies. This verifies that the proposed model can deliver high and stable returns when deployed as an investment strategy in stock market trading.

(3) Broad theoretical value and application prospects

This study not only provides a novel and efficient solution for financial time-series forecasting, but its modular design paradigm also exhibits broad applicability, offering valuable insights for tackling complex time-series forecasting tasks in other domains, such as meteorology and energy-load forecasting.

However, this study also has certain limitations. Future research can be expanded in the following directions:

(1) Future work can focus on model lightweighting, integration of multi-source data (e.g., news sentiment, macroeconomic indicators), and the development of online learning mechanisms to further enhance practical utility and applicability.

- (2) In this study, stock selection was based on time-series properties, risk-return profiles, and volatility considerations, with three representative stocks chosen from the financial, infrastructure, and consumer manufacturing sectors for validation. Future studies could test the model's generalization and robustness by applying it to different markets (e.g., U.S. stocks, Hong Kong stocks, cryptocurrencies) and under varying market volatility regimes.
- (3) Dynamic weight adjustment and memory mechanisms could be incorporated into the model framework to enable self-learning and self-regulation capabilities, allowing the model to automatically optimize the balance between exploration and exploitation during training based on convergence trends.
- (4) Further exploration could extend this methodology to other domains such as multi-region power grid-load forecasting and renewable energy generation prediction, broadening its applicability and impact.

Author Contributions: Conceptualization, H.W.; Methodology, X.H.; Software, X.H.; Formal analysis, X.H.; Resources, H.W.; Writing—original draft, X.H.; Writing—review & editing, H.W.; Funding acquisition, H.W. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Basic Scientific Research Project of the Department of Education of Liaoning Province (Project title: Generative AI in Multimodal Risk Scenario Simulation and Dynamic Emergency Decision-Making, Grant No. LJ212510149011).

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Ye, P.; Zhang, H.; Zhou, X. CNN-CBAM-LSTM: Enhancing Stock Return Prediction Through Long and Short Information Mining in Stock Prediction. *Mathematics* **2024**, *12*, 3738. [\[CrossRef\]](#)
2. Tamilselvi, C.; Yeasin, M.; Paul, R.K.; Paul, A.K. Can denoising enhance prediction accuracy of learning models? A case of wavelet decomposition approach. *Forecasting* **2024**, *6*, 81–99. [\[CrossRef\]](#)
3. Zhang, J.; Liu, H.; Bai, W.; Li, X. A hybrid approach of wavelet transform, ARIMA and LSTM model for the share price index futures forecasting. *N. Am. J. Econ. Financ.* **2024**, *69*, 102022. [\[CrossRef\]](#)
4. Wang, Y.; Zhang, D.; Li, M.; Zhang, X. Study on power prediction of photovoltaic power generation based on NGO-LSTM. In *2024 8th International Conference on Electrical, Mechanical and Computer Engineering (ICEMCE), Xian, China, 25–27 October 2024*; IEEE: Piscataway, NJ, USA, 2024; pp. 224–228.
5. Huiyong, W.; Wang, Z. Stock market forecasting research based on GA-WOA-LSTM. *PLoS ONE* **2025**, *20*, e0330324. [\[CrossRef\]](#)
6. Liu, Z.; Liu, X.; Zhang, H. Short-Term Wind Power Prediction Based on Improved SAO-Optimized LSTM. *Processes* **2025**, *13*, 2192. [\[CrossRef\]](#)
7. Lu, W.; Li, J.; Wang, J.; Qin, L. A CNN-BiLSTM-AM method for stock price prediction. *Neural Comput. Appl.* **2021**, *33*, 4741–4753.
8. Krizhevsky, A.; Sutskever, I.; Hinton, G.E. ImageNet classification with deep convolutional neural networks. *Commun. ACM* **2017**, *60*, 84–90. [\[CrossRef\]](#)
9. Ashtiani, M.N.; Raahemi, B. News-based intelligent prediction of financial markets using text mining and machine learning: A systematic literature review. *Expert Syst. Appl.* **2023**, *217*, 119509. [\[CrossRef\]](#)
10. D'Amico, S.; Mercorio, F.; Nobani, N.; Sperli, G.; Ventre, C. Learning across modalities: A systematic survey of multimodal models for financial analysis. *Inf. Fusion* **2026**, *133*, 104249. [\[CrossRef\]](#)
11. Beniwal, M.; Singh, A.; Kumar, N. Forecasting multistep daily stock prices for long-term investment decisions: A study of deep learning models on global indices. *Eng. Appl. Artif. Intell.* **2024**, *129*, 107617. [\[CrossRef\]](#)
12. Livieris, I.E. C-kan: A new approach for integrating convolutional layers with kolmogorov–arnold networks for time-series forecasting. *Mathematics* **2024**, *12*, 3022. [\[CrossRef\]](#)
13. Zhang, Y.; Cui, L.; Yan, W. Integrating Kolmogorov–Arnold Networks with Time Series Prediction Framework in Electricity Demand Forecasting. *Energies* **2025**, *18*, 1365. [\[CrossRef\]](#)
14. He, Z.; Guo, Z. Time Series Data Prediction Using CNN-BiLSTM based Attention-KAN Model. In *2024 IEEE Smart World Congress (SWC), Nadi, Fiji, 2–7 December 2024*; IEEE: Piscataway, NJ, USA, 2024; pp. 2329–2335.

15. Jamali, A.; Roy, S.K.; Hong, D.; Lu, B.; Ghamisi, P. How to learn more? Exploring Kolmogorov–Arnold networks for hyperspectral image classification. *Remote Sens.* **2024**, *16*, 4015. [\[CrossRef\]](#)
16. Li, Y.; Wu, H.; Liang, C.; Bu, C.; Jin, H.; Chen, L. An Attention-Guided Improved Decomposition-Reconstruction Model for Stock Market Prediction. *IEEE Access* **2025**, *13*, 167893–167911. [\[CrossRef\]](#)
17. Dai, Z.; Tang, R. The impact of oil price shocks on systematic risk of G7 stock markets. *Expert Syst. Appl.* **2024**, *248*, 123408. [\[CrossRef\]](#)
18. Sezer, O.B.; Gudelek, M.U.; Ozbayoglu, A.M. Financial time series forecasting with deep learning: A systematic literature review: 2005–2019. *Appl. Soft Comput.* **2020**, *90*, 106181. [\[CrossRef\]](#)
19. Chen, L.; Fan, X. TIC-FusionNet: A multimodal deep learning framework with temporal decomposition and attention-based fusion for time series forecasting. *PLoS ONE* **2025**, *20*, e0333379. [\[CrossRef\]](#) [\[PubMed\]](#)
20. Sukma, N.; Namahoot, C.S. Enhancing trading strategies: A multi-indicator analysis for profitable algorithmic trading. *Comput. Econ.* **2025**, *65*, 3807–3840.
21. Yi, S.; Chi, J.; Shi, Y.; Zhang, C. Robust stock trend prediction via volatility detection and hierarchical multi-relational hypergraph attention. *Knowl.-Based Syst.* **2025**, *287*, 114283. [\[CrossRef\]](#)
22. Wang, C.H.; Lin, C. Stock conditional drawdown at risk portfolio optimization based on gated bidirectional temporal convolution and discrete cosine graph neural networks on hypervariable graphs. *Knowl.-Based Syst.* **2025**, *288*, 114456. [\[CrossRef\]](#)
23. Teng, X.; Zhang, X.; Luo, Z. Multi-scale local cues and hierarchical attention-based LSTM for stock price trend prediction. *Neurocomputing* **2022**, *505*, 92–100. [\[CrossRef\]](#)
24. Teng, X.; Zhang, L.; Gao, P.; Yu, C.; Sun, S. BERT-Driven stock price trend prediction utilizing tokenized stock data and multi-step optimization approach. *Appl. Soft Comput.* **2025**, *170*, 112627. [\[CrossRef\]](#)
25. Yang, Z.; Zhao, T.; Wang, S.; Li, X. MDF-DMC: A stock prediction model combining multi-view stock data features with dynamic market correlation information. *Expert Syst. Appl.* **2024**, *238*, 122134. [\[CrossRef\]](#)
26. Ge, Q. Enhancing stock market Forecasting: A hybrid model for accurate prediction of S&P 500 and CSI 300 future prices. *Expert Syst. Appl.* **2025**, *260*, 125380. [\[CrossRef\]](#)
27. Alzaman, C. Optimizing portfolio selection through stock ranking and matching: A reinforcement learning approach. *Expert Syst. Appl.* **2025**, *269*, 126430. [\[CrossRef\]](#)
28. Morlet, J.; Arens, G.; Fargeau, E.; Giard, D. Wave propagation and sampling theory—Part II: Sampling theory and complex waves. *Geophysics* **1982**, *47*, 222–236. [\[CrossRef\]](#)
29. Wang, L.; Yao, W.; Chen, C.; Yang, H. Driving behavior recognition algorithm combining attention mechanism and lightweight network. *Entropy* **2022**, *24*, 984. [\[CrossRef\]](#) [\[PubMed\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.